Discovery Lab

Exam 1 Revision

Lower Secondary Stage 2 · Science · Lessons 1 to 6

Life processes, tissues, cells and organ systems, mixtures, evaporation and distillation, and chromatography.



Lesson 1: Life Processes and Organs

Lesson 2: Animal and Plant Tissues

Lesson 3: Cells and Organ Systems

Lesson 4: Mixtures and Solutions

Lesson 5: Evaporation and Distillation

Lesson 6: Chromatography

Name: _____ Class: _____ Date: _____

LESSON 1

1 Life Processes and Organs

What it means to be alive and which organs keep you going.

Key ideas

Something is alive only if it does **all eight** life processes: **M**ovement, **R**espiration, **S**ensitivity, **C**ontrol of conditions, **G**rowth, **R**eproduction, **E**xcretion, **N**utrition.

Respiration is not breathing: it is the chemical process in cells that **releases energy from food**.

Photosynthesis is not one of the eight: it is how a plant makes food, part of **nutrition**.

An **organ** is a group of tissues that does one job. Plants have organs too (roots, leaves).

Organ	Job
Heart	Pumps blood around the body
Lungs	Take oxygen from the air
Brain	Controls thoughts and body functions
Liver	Removes toxins from the blood
Roots	Absorb water from the soil
Leaves	Carry out photosynthesis

DID YOU KNOW?

Doctors can transplant organs to save lives, and scientists are growing artificial organs.

Check yourself

- [ ] Name the eight life processes.
- [ ] Is respiration the same as breathing?
- [ ] Is photosynthesis a life process?
- [ ] What is an organ?
- [ ] Do plants have organs?

REVISION TASK

Make a flashcard for each of the eight life processes with an example from a living thing.

LESSON 2

2 Animal and Plant Tissues

Similar cells working together form tissues.

Key ideas

Animal tissues: muscle (movement), nerve (messages to and from the brain), epithelial (covers and protects), connective (structure and support, including **blood**).

Plant tissues: **xylem** carries water up, **phloem** carries food (sugars), root tissue absorbs water.

An organ is made of **more than one tissue**.

Cells → tissues → organs → organ systems.

Tissue	Found in	Job
Muscle	Animals	Helps movement
Nerve	Animals	Sends messages
Xylem	Plants	Carries water up
Phloem	Plants	Carries food

DID YOU KNOW?

Muscle tissue is made of long bundles of fibres that contract to cause movement.

Check yourself

- [ ] What is a tissue?
- [ ] What does xylem do?
- [ ] What does phloem do?
- [ ] Why is blood a tissue?
- [ ] Can an organ have more than one tissue?

REVISION TASK

Draw the chain cells → tissues → organs → systems with one example of each.

LESSON 3

3 Cells and Organ Systems

Cells make tissues, tissues make organs and organs make systems.

Key ideas

In plant **and** animal cells: **nucleus** (controls the cell), **cell membrane** (controls what enters and exits), **cytoplasm** (chemical reactions), **mitochondria** (release energy).

Only in **plant cells**: **cell wall** (strength), **chloroplasts** (photosynthesis), **large vacuole** (stores water and helps shape).

Kidneys filter blood and remove waste (urea) and extra water as **urine**. If they fail, a person may need **dialysis** or a **transplant**.

System	Job	Organs
Digestive	Breaks down food	Stomach, intestines
Respiratory	Takes in oxygen	Lungs, trachea
Circulatory	Moves substances	Heart, blood vessels
Excretory	Removes waste	Kidneys, bladder
Nervous	Control and messages	Brain, nerves

DID YOU KNOW?

Only plant cells have a cell wall, chloroplasts and a large vacuole.

Check yourself

- [ ] Which cell parts are only in plant cells?
- [ ] What does the nucleus do?
- [ ] What do mitochondria do?
- [ ] What do the kidneys do?
- [ ] What can help if the kidneys fail?

REVISION TASK

Draw a plant cell and an animal cell and label every part.

LESSON 4

4 Mixtures and Solutions

Substances mixed together but not chemically joined.

Key ideas

A **mixture** has two or more substances **not chemically joined**. It can be separated **physically**.

A **solution**: a substance (**solute**) dissolves in another (**solvent**). Salt in water: salt is the solute, water is the solvent.

The solute seems to vanish but is still there and can be recovered. A **cloudy** mixture has undissolved solid.

Filtration separates solids from liquids. **Sieving** separates big pieces from small. A **magnet** separates magnetic materials.

Method	Separates
Filtration	Solids from liquids
Evaporation	Recovers the solute from a solution
Sieving	Large solids from small solids
Magnet	Magnetic from non-magnetic materials

DID YOU KNOW?

Muddy water is a mixture: the soil is suspended and filtration holds it back.

Check yourself

- [ ] What is a mixture?
- [ ] What are solute and solvent?
- [ ] Why does filtering not remove dissolved salt?
- [ ] Which method separates iron filings from sand?
- [ ] What does a cloudy mixture show?

REVISION TASK

Make a table of four separation methods and an example of each.

LESSON 5

5 Evaporation and Distillation

Recover the solute, or recover both parts.

Key ideas

Evaporation: particles gain heat energy and become gas. It separates a **dissolved solid** from a liquid. The solute is kept; the solvent is lost as vapour.

Distillation: the liquid is heated and evaporates, then the vapour is **cooled and condensed** in a separate container. It keeps **both** the solute and the solvent.

Astronauts recycle urine into drinking water by distillation.

	Evaporation	Distillation
Heat	Yes	Yes
Cooling	No	Yes (condenser)
Solute kept?	Yes	Yes (left in flask)
Solvent kept?	No: lost as vapour	Yes: collected

DID YOU KNOW?

Salt ponds use the Sun to evaporate the water and leave salt behind.

Check yourself

- [ ] What is the difference between evaporation and distillation?
- [ ] Where does the vapour go in distillation?
- [ ] How would you get salt from salty water?
- [ ] Why is filtering a bad way to remove salt?
- [ ] Name a real use of distillation.

REVISION TASK

Draw a flow chart that follows the salt and the water in evaporation and in distillation.

LESSON 6

6 Chromatography

Separate the colours hidden inside an ink.

Key ideas

Chromatography separates a **mixture of coloured substances** (dyes).

Stationary phase = the paper. **Mobile phase** = the solvent travelling up the paper. The result is a **chromatogram**.

Dyes have different **solubility**. A more soluble dye travels **further**.

Start the spot **above** the solvent level. A pure dye gives **one spot**. Uses: forensics and food colouring checks.

Term	Meaning
Stationary phase	The paper
Mobile phase	The solvent
Chromatogram	The finished paper
More soluble dye	Travels further

DID YOU KNOW?

Black ink is not one dye: it is a mixture of several.

Check yourself

- [ ] What does chromatography separate?
- [ ] What is the stationary phase?
- [ ] Why do some dyes travel further?
- [ ] Why start the spot above the solvent?
- [ ] Give two real uses.

REVISION TASK

Draw a chromatogram of a black ink that separates into three dyes and explain why.

FINAL PRACTICE

Final practice

Tick each box when you can answer without looking.

Final practice

- [ ] Explain how tissues, organs and systems are related.
- [ ] Which cell parts are only in plant cells?
- [ ] How do evaporation and distillation differ?
- [ ] Why can chromatography separate an ink?
- [ ] Which method separates iron filings from sand?

CHALLENGE

Write a short paragraph describing how you would separate a mixture of sand, salt and water.

Revision checklist

Revision task	Done?
Reviewed the Canvas slides for Lessons 1–6	☐
Answered every "Check yourself" question	☐
Drew cells, tables and flow charts	☐
Practised the long answers on the website	☐
Played the 3D cell and the labs	☐
Asked for help with anything unclear	☐

Try it live · Practise with 3D labs, games and a practice exam at **morehumaned.com/discovery-lab**

ANSWER KEY

Answer key

Check your own answers against these. If one is hard, go back to the lesson.

Lesson 1: Life Processes and Organs

1. Movement, respiration, sensitivity, control of conditions, growth, reproduction, excretion, nutrition.
2. No: respiration is the chemical process that releases energy from food.
3. Not on its own: it is part of nutrition.
4. A group of tissues that work together to do a job.
5. Yes: roots and leaves.

Lesson 2: Animal and Plant Tissues

1. A group of similar cells working together.
2. Carries water up the plant.
3. Carries food (sugars).
4. It is made of cells that carry and support the body.
5. Yes: organs are made of several tissues.

Lesson 3: Cells and Organ Systems

1. Cell wall, chloroplasts and a large vacuole.
2. It controls the cell.
3. They release energy by respiration.
4. They filter blood and remove waste and extra water.
5. Dialysis or a kidney transplant.

Lesson 4: Mixtures and Solutions

1. Two or more substances not chemically joined.
2. The solute dissolves in the solvent.
3. Dissolved particles are too small to be held back by the filter.
4. A magnet.
5. Solid is suspended and has not dissolved.

Lesson 5: Evaporation and Distillation

1. Distillation also cools and collects the solvent.
2. To a condenser and then a separate container.
3. Evaporate the water.
4. The salt is dissolved and passes through.
5. Recycling water on the space station, or purifying water.

Lesson 6: Chromatography

1. A mixture of coloured substances.
2. The paper.
3. They are more soluble in the solvent.
4. So the dyes do not dissolve into the solvent in the beaker.
5. Forensic ink comparison and checking food colourings.

Final practice

1. Cells form tissues, tissues form organs, organs form systems.
2. Cell wall, chloroplasts and large vacuole.
3. Distillation also cools and collects the solvent.
4. Dyes differ in solubility and travel different distances.
5. A magnet.

Discovery Lab

You are going to do great! Use this booklet to plan your study time, then bring it all together.

Mr Guevara · morehumaned.com/discovery-lab